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ENERGY (CONVENTIONAL AND NON-CONVENTIONAL)

INTRODUCTION

Energy is a very common word frequently used in our day-to-day life. Energy is defined as the ability to do work. We require energy for all types of activities including the activities within our body, with our body or with other bodies. When we say a body has energy, it means that it is capable of doing work. Look around you will find countless examples where energy is used to do work. An engine uses energy of its fuel to move a car along. A battery stores the energy needed to switch on the radio or tape recorder. The heavy flow of water can break the banks of rivers as it also has energy in it. Similarly, the wind also carries enough energy to shake trees.

Importance of Energy in our Life:

Energy plays a very important role in our lives, providing comfort, increasing productivity and allowing us to live the way we want to. Since the beginning of mankind, we have made use of wood, water, and fossil fuels as a means of heating and making machines work. Almost for all types of activities, we rely on one or another form of energy. Amount of energy used by a society is an indicator of its economic growth and development. Without energy even our body would be unable to perform basic functions like respiratory, circulatory, or digestive functions to name a few. Plants would also be unable to complete the process of converting Carbon dioxide, water and minerals into food without the light from the Sun. Almost all the machines used for the production and manufacture of different types of items would be unable to operate without the use of a source of electrical energy. Almost everything we see around us, the clothes we wear, the food we eat, the houses we live in, the paper we write on, the vehicles we drive, all need energy to be created or transformed from some natural resource to the final product. Nowadays, the electrical energy has become so important that almost in all walks of life electricity is required. For example, all electrical appliances in our homes and at our workplace require electricity. All the industries and factories run on electricity.

Various forms of Energy:

In our daily life we use different forms of energy such as heat energy, light energy, mechanical energy, electrical energy, chemical energy, and sound energy. The most common forms of energy are heat, light and electricity. We use all these forms of energies for different types of work. As per requirement, one form of energy can be converted into another form of energy by using specific types of devices or processes. We get energy for our daily use from different sources.

Different Sources of Energy:

In simple terms we can say that anything out of which usable energy can be extracted is a source of energy. There is a variety of sources that provide us energy for different purposes. You must be familiar with coal, petrol, diesel kerosene and natural gas. Similarly, you must have also heard about hydroelectric power, wind mills, solar panels, biomass etc. It can be easily seen that some of the energy sources can be replenished in a short period of time. Such energy sources are referred to as "renewable" energy sources, whereas the energy sources that we are using up and cannot be generated in a short period of time are called non-renewable energy sources. Thus, all the sources of energy can be divided into two categories: renewable sources and non-renewable sources of energy.

CONVENTIONAL AND NON-CONVENTIONAL SOURCES

Conventional Sources:

Coal

Coal is a fossil fuel. It is a flammable, black or brown sedimentary rock and is mainly composed of carbon. It is the altered remains of prehistoric vegetation that originally accumulated in swamps and peat bogs. The dense forest plants were converted into coal due to intense pressure and heat inside the earth by the process of carbonization. Most of the coal resources of the world were formed during the carboniferous period (280 to 350 million years ago). The quality of the coal is determined by its carbon content. The following types of coal have been identified on the basis of their physical properties. They are,

(i) **Peat** is the first stage of transformation of wood into coal and it has only 30 to 35% of carbon.

(ii) **Lignite or Brown coal** is the inferior quality and contains 35-45% carbon

(iii) **Bituminous or coking coal** is the second-best variety of coal and contains 70-90% of carbon. It is the most widely spread and most widely used variety of coal. It is the most popular coal in commercial use.

(iv)**Anthracite** is the best quality coal, which contains more than 95% of carbon. It is very hard but emits very less smoke and leaves very less ash. However, its deposits are limited.

Production and world distribution of Coal:

Coal reserves are found in more than 70 countries of the world but the major coal reserves occur in the USA, Russia, China and South Africa. China is the largest producer of steam coal in the world followed by India. The other leading producers of steam coal are USA, Indonesia, and South Africa etc.

Steam coal – It is used for producing steam and it has high Sulphur content

China was the largest producer of coking coal in the world in 2016 followed by Australia. The other leading producers of coking coal are Russia, India and USA.

Major coal mining centres:

Country	**Mining centres**
China	Shansi, Shantung, Fushun, Shenyang, etc.
India	Ranigunj (Oldest coal field in India)-West Bengal Bokaro,Jharia-Jharkhand Korba-Chhatisgarh Singareni-Andhra Pradesh
U.S.A	Arkansas, Colorodo,

	Illionois, Indiana, Michigan etc.
Australia	Bowen Basin, Brisbane, Canberra, Sydney, New-castle, Tasmania, etc.
Russia	Moscow-Tula region, Chokot Basin, Ob basin, etc.

Petroleum (or) Mineral oil:

Petroleum is a mineral that exists under the surface of the earth in liquid, solid and gaseous forms. Liquid petroleum may be in the form of crude oil. The solid form may be mineral waxes or asphalts. The gaseous form is natural gas. It is a main source of energy in the World due to its multiple uses. The human activities are directly or indirectly depending on the use of petroleum or its sub products.

Formation and occurrence of mineral oil

It is formed by slow chemical and bio chemical decomposition of the remains of organic matter in sedimentary rocks. It is found in the pores of the sedimentary rocks. Oil is lighter than water hence, floats over water. Drilling of oil wells is the hole drilled in the earth's crust and when it reaches the rock cap, the natural gas comes out first with a great pressure. When the pressure of gas subsides, petroleum starts flowing out when the pressure of natural gas is released.

Petroleum reserves of the world

The west Asia or Middle East is having the largest petroleum reserves, which is about 60% of the world's oil reserve. The total estimated world's oil reserves in 2008 were 1,243 (109 bbl). Saudi Arabia, Canada, Iran, Iraq and Kuwait have large reserves of petroleum.

Major Petroleum production centres:

Country	Production centres
Saudi Arabia	Ghawar, Abquiaq, Abuhadriya, etc.
Russia	Volga- Caspian region, Kamchatka- Sakhalin region, Ob – Lena basin.
U.S.A	Tennessee- New york, ohio, Indiana, Pennsylvania, Texas, Mississippi, Gulf of California, etc.
Iraq	Kirkuk, Mosul, Daura, etc.
China	Taching, Chinchou, yemen, South china sea, etc.

Production and world distribution of petroleum

The petroleum producing countries of the world can be grouped in to five geographical regions:

1. West Asia (or) Middle East region
2. American region
3. Russian region
4. East & South Asian region and
5. African region

Saudi Arabia is the largest oil producer of the world with 13.62% of the world output of oil. Russia is the second largest producer in the world. India is placed at 24th position in petroleum production in the world. The distribution of oil is naturally uneven; Middle East contains 60% of global reserves and rest of the world only 40%.

Natural Gas

It is the cheapest source of energy. It is found along with or without petroleum. It is considered as an environment friendly fuel because of its low carbon dioxide emissions. Therefore, this is the only fuel for the present century and it is also called **green energy.** A powerful odorant, ethanethiol is added, so that leaks can be detected easily. It is prepared by refining petroleum or wet natural gas.

Natural gas reserves and Production

The known natural gas reserves in the world is about 6254 trillion cubic feet. Most of these reserves are found in Russia, Iran, Qatar, UAE, Saudi Arabia, USA etc. USA has the largest reserve and is the leading producer of natural gas in the world followed by Russia. India is the 28th producer of natural gas in the world. It is widely used as a fuel in industries and domestic cooking purposes. Petrochemical industries use it as fuel and raw material. It is also used in chemical industries, artificial rubber, plastic, fertilizers, ink, and carbon and as artificial lighting.

Nuclear Energy

It is commonly said, this energy holds the key of future. Energy contained within the nucleus of an atom is called nuclear energy. Heavy metals like Uranium, Thorium, Radium, Plutonium and Lithium are the main sources of nuclear energy. However, Uranium is the most important source of nuclear energy. The nuclear energy production was started first in USA in 1950.

Nuclear energy now provides about 11% of the World's electricity. At present there are more than 450 operable fission reactors in the world. The world's first commercial nuclear power station Calder Hall at Wind scale, England was opened in 1956.

Renewable sources of Energy:

All regions of the world are facing the twin problems of fast increasing demand for energy and limited supplies and rapidly depleting conventional sources of energy. Under these circumstances, non-conventional sources of energy are getting more importance. These sources are renewable, clean and non-polluting. They are solar, wind, geothermal, wave, tidal energy, bio-gas etc.

Hydel Power

Hydro electricity is produced by using the potential energy of water falling from a certain height. The falling water spins the turbine blades and energy is produced. It is a clean ecofriendly and renewable source of energy. It contributes nearly 7% of the world electricity production. China has the largest potential followed by Brazil, Indonesia, Canada and Zaire. China is the largest producer of Hydroelectricity in the world, followed by Canada.

Solar energy

It is based on mechanical conversion of solar energy into electricity. It is available in abundance but only in the recent period it gets more importance due to technological development. Solar energy is used for various purposes. USA is the major producer of solar cells at present. It is simply the energy provided by the sun, which makes production of solar electricity possible.

Solar power in India is a fast-developing industry. The country's solar installed capacity reached 26 GW as of 30 September 2018. India expanded its solar generation capacity 8 times from 2,650 MW on 26 May 2014 to over 20 GW as on 31 January 2018.

The country added 3 GW of solar capacity in 2015-2016, 5 GW in 2016-2017 and over 10 GW in 2017-2018, with the average current price of solar electricity dropping to 18% below the average price of its coal-fired counterpart.

Wind Energy

The wind is a clean, free and readily available renewable energy source. Wind turbines are capturing the wind's power and converting it to electricity. Wind power has become a pillar in their strategies to phase out fossil and nuclear energy. Wind energy is now the second fastest growing source of electricity in the world. It fulfils about 5% of world's electricity demand. The world's largest wind farm is in Altamont pass in California. India is emerging as a major wind power producer of world.

The important wind farms in India –

The largest wind farms in India are Muppandal in Kanyakumari District of Tamil Nadu and Jaisalmer wind park in Rajasthan. They are the first and second largest wind farms of India. Based on the location of its generation it is classified into

1. Onshore wind energy and
2. Offshore wind energy

1.Onshore wind energy- Energy generated from the plants located on the land is known as onshore wind energy. Onshore wind has the advantage of being one of the most affordable renewable energy sources. It is cheaper than any other renewable source of energy but it requires more area to install than any other energy.

2. Offshore wind energy –It refers to the use of wind farms developed in seas and oceans. The largest offshore wind farms are currently in the U.K and Germany. These two countries installed 2/3 capacity. London Array is the largest offshore wind farm in the world. The first offshore wind farm is planned near Dhanuskodi in Tamil Nadu.

Tidal energy

It is a renewable energy powered by the natural raise and fall of ocean water. Its production is very small. The first tidal power station was located in La Rance in France. The largest tidal power station is at Sihwa Lake in South Korea and it is the largest tidal power producer in the world. There are three different category of sources from which the tidal energy is generated. The sources are tidal streams, barrages and tidal lagoons. India's first attempt to harness tidal power for generating electricity would be in the form of a 3MW plant at the Durgaduani creek in sunderbans delta of West Bengal. The Gulf of Kutch and Cambay in Gujarat and the Ganges delta in sunderbans, the world's largest mangrove, are the 3 sites identified as potential areas for tidal power generation in India.

Geo Thermal Energy

Geo thermal energy is derived from the natural heat of the earth. The United States is the world's largest producer, and the largest geothermal development in the world is the Geysers north of San Francisco in California, the U.S.

In India, exploration and study of geothermal fields started in 1970. The GSI (Geological Survey of India) has identified 350 geothermal energy locations in the country. The most promising of these is in Puga valley of Ladakh. The estimated potential for geothermal energy in India is about 10000 MW. There are seven geothermal provinces in India: the Himalayas, Sohana, West coast, Cambay, Son-Narmada-Tapti (SONATA), Godavari, and Mahanadi.

Questions:

1. Explain the importance of Energy in our life and Mention the various forms of Energy.
2. Mention some of the Energy Resources and its distributions in India.
3. Describe about the Conventional Energy Resources in India.